

Intro

I was tasked with evaluating a simple circuit with two resistors in series with each other and a 10 V voltage source. I computed the power consumption of R_1 by first calculating the current of the mesh, so I knew

$$V = IR$$

So

$$I = \frac{V}{R}$$

And I knew

$$R_{EQ} = R_1 + R_2$$

for resistors in series

Thus

$$I = \frac{V}{R_{EQ}} = \frac{V}{R_1 + R_2}$$

I also knew that

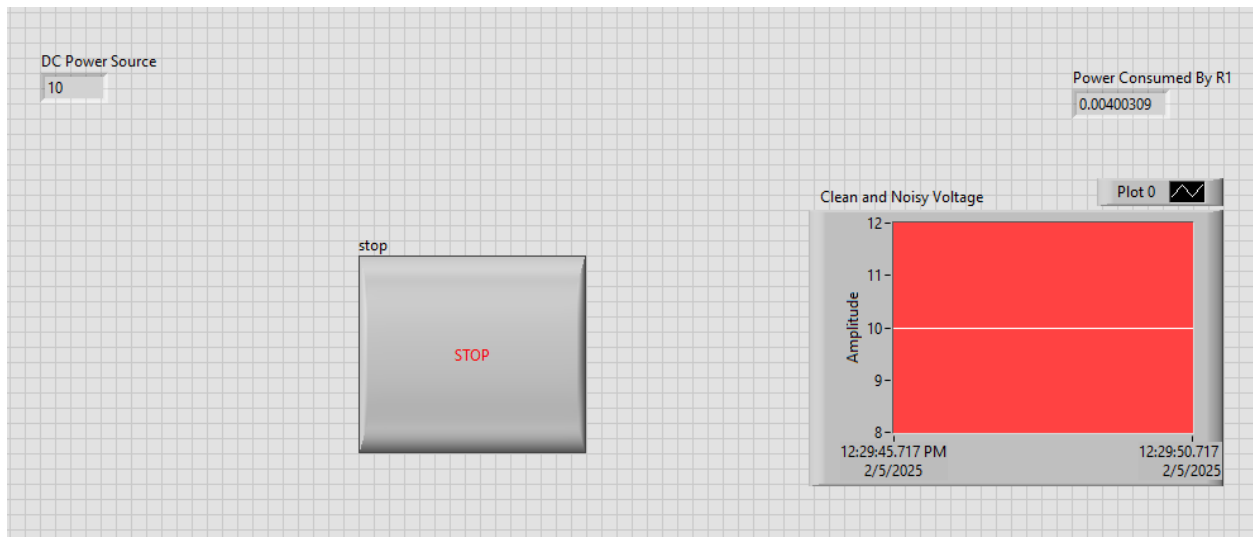
$$P = VI$$

So

$$P = I^2R$$

Thus, I was now able to calculate the power consumption of R_1 using this formulation.

Front Panel:

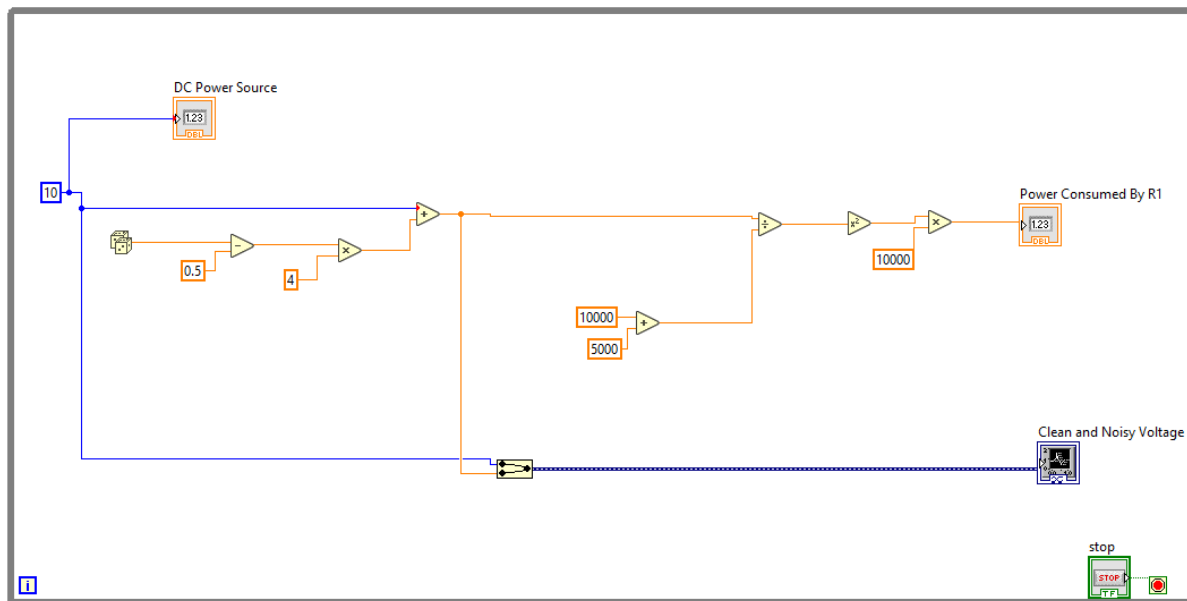


Since I knew the power source should be 10 DCV I created it as a constant in the block diagram then displayed it in the front panel using a numeric indicator.

Then I created a noise generator using rnum from 0 to 1, subtracting to make it -0.5 to 0.5 , and multiplying to make it -2 to 2 .

To plot the clean and noisy voltage on the same graph I used a merge signal inputted into a waveform chart. I made sure to get the clean signal from the numeric constant directly and to get the noisy signal from the combination of the noise and the clean signal. This was beneficial for checking that my noise generator was right as well.

Block Diagram:



Using the same line of logic from the Intro section, after I added the noise to the voltage I performed the necessary operations.

I represented R_1 and R_2 as constants added together to be used in

$$I = \frac{V}{R_{EQ}}$$

Now the output of the division operation was the current of the mesh and thereby R_1 .

Then following the same formula as before I squared the current and divided by R_1 to get the power dissipation of R_1 and displayed it to another number indicator.

I also added a while loop since specifications wanted it to run as long as user wants, stopping with the STOP button.